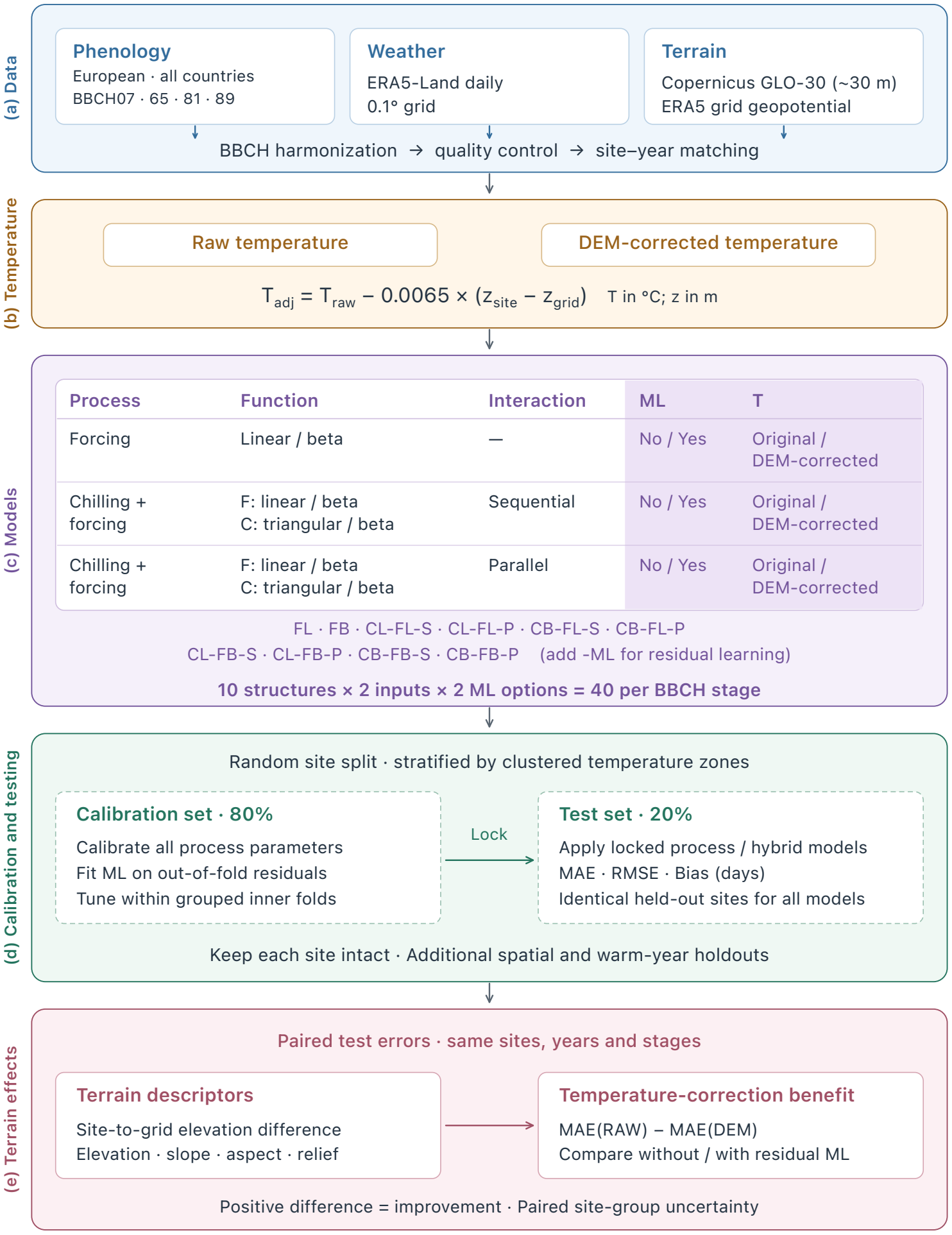
(a) Data
Phenology
European · all countries
BBCH07 · 65 · 81 · 89
Weather
ERA5-Land daily
0.1° grid
Terrain
Copernicus GLO-30 (~30 m)
ERA5 grid geopotential
BBCH harmonization → quality control → site–year matching
(b) Temperature
Raw temperature
DEM-corrected temperature
$T_{adj} = T_{raw} - 0.0065 \times (z_{site} - z_{grid})$ T in °C; z in m
(c) Models
Process
Function
Interaction
ML
T
Forcing
Linear / beta
—
No / Yes
Original / DEM-corrected
Chilling + forcing
F: linear / beta
C: triangular / beta
Sequential
No / Yes
Original / DEM-corrected
Chilling + forcing
F: linear / beta
C: triangular / beta
Parallel
No / Yes
Original / DEM-corrected
FL · FB · CL-FL-S · CL-FL-P · CB-FL-S · CB-FL-P
CL-FB-S · CL-FB-P · CB-FB-S · CB-FB-P (add -ML for residual learning)
10 structures × 2 inputs × 2 ML options = 40 per BBCH stage
(d) Calibration and testing
Random site split · stratified by clustered temperature zones
Calibration set · 80%
Calibrate all process parameters
Fit ML on out-of-fold residuals
Tune within grouped inner folds
Lock
Test set · 20%
Apply locked process / hybrid models
MAE · RMSE · Bias (days)
Identical held-out sites for all models
Keep each site intact · Additional spatial and warm-year holdouts
(e) Terrain effects
Paired test errors · same sites, years and stages
Terrain descriptors
Site-to-grid elevation difference
Elevation · slope · aspect · relief
Temperature-correction benefit
MAE(RAW) – MAE(DEM)
Compare without / with residual ML
Positive difference = improvement · Paired site-group uncertainty